

SOFTWARE TESTING AUTOMATION

The **Software Testing (Manual, Automation, SQL)** course has been developed in collaboration with hundreds of hiring companies in India to prepare fresh graduates for roles as Software Testing Engineers or Quality Analysts. This course covers Manual Testing, Automation Testing, and SQL, and is designed to assist candidates in preparing for the International Software Testing Qualifications Board (ISTQB) examination



JAVA

1. INTRODUCTION OF JAVA

1.1.1. Platform independent

1.1.2. Open source

1.1.3. Multi-threading

1.1.4. More secure

1.1.5. Portable

2. CORE JAVA TOPICS

2.1.1. OOPS concept

2.1.2. Control statement and looping

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2.1.3. Arrays

2.1.4. String

2.1.5. Collections

2.1.6. Exceptions

2.1.7. Constructor

3. INTRODUCTION OF OOPS

3.1.1. Class

3.1.2. Object

3.1.3. Method

3.1.4. Encapsulation

3.1.5. Inheritance

3.1.6. Polymorphism

3.1.7. Abstraction



4. ENCAPSULATION

5. INHERITANCE

5.1.1. Single inheritance

5.1.2. Multilevel inheritance

5.1.3. Multiple inheritance

5.1.4. Hybrid inheritance

5.1.5. Hierarchical inheritance

6. ACCESS SPECIFIER

6.1.1. Public

6.1.2. Private

6.1.3. Protected

6.1.4. Default7. **DATA TYPES AND WRAPPER CLASS**

7.1.1. Data types

7.1.2. Wrapper class

8. CONTROL STATEMENT

8.1.1. If

8.1.2. If-else

8.1.3. If-else-if

9. LOOPING

9.1.1. For

9.1.2. While

9.1.3. Do-while

10. POLYMORPHISM

10.1.1. Method overloading

10.1.2. Method overriding

11. ABSTRACTION

11.1.1. Abstract class

11.1.2. Interface

12. ARRAYS



13. STRING

13.1.1. String functions

13.1.2. Literal string

13.1.3. Non-literal string

13.1.4. Mutable string

13.1.5. Immutable string

14. COLLECTIONS

14.1.1. List

14.1.2. Set

14.1.3. Map

15. EXCEPTION

15.1.1. Unchecked exception

15.1.2. Checked exception

15.1.3. Exception handling

16. CONSTRUCTOR

17. CORE JAVA INTERVIEW QUESTIONS AND ANSWERS

SELENIUM AUTOMATION

1. INTRODUCTION OF AUTOMATION TESTING

2. BROWSER LAUNCHING

3. WEB DRIVER METHODS

4. NAVIGATION COMMANDS

5. LOCATORS

6. WEBELEMENT METHODS

7. RADIO BUTTON

8. HANDLING MULTIPLE WEBELEMENTS AND AUTO SUGGESTION

9. CHECK BOX

10. DROP DOWN

11. WEB TABLE

12. JAVASCRIPT EXECUTOR

13. SCROLL UP AND SCROLL DOWN

14. SCREENSHOT

15. MOUSE HOVER ACTION

16. DRAG & DROP AND RIGHT CLICK

17. WINDOW HANDLING

18. FRAME HANDLING

19. ALERTS

20. WAIT

21. FILE UPLOADING USING ROBOT CLASS AND SENDKEYS

22. BROKEN LINKS

1. Introduction to Software Testing

1. What is Software Testing
2. What is Quality
3. Importance of Testing
4. Tools Used for Testing
5. Manual and Automation Testing
6. Principles of Software Testing

2. Software Development Life Cycle (SDLC)

1. SDLC and SDLC Phases
2. Requirements and Analysis
3. Design
4. Development
5. Testing
6. Deployment
7. Operation and Maintenance



3. SDLC Models

1. Waterfall Model
2. V Model
3. Spiral Model
4. Iterative Model

5. Agile Model

4. Development Architectures

1. One-Tier Architectures
2. Two-Tier Architectures
3. Three-Tier Architecture
4. N-Tier Architecture

5. Software Testing Life Cycle (STLC)

1. What is STLC?
2. Entry and Exit Criteria
3. STLC Phases
 - Requirement Analysis
 - Test Planning
 - Test Design
 - Test Execution
 - Sign Off
4. Test Scenario Preparation
5. Test Case Preparation
6. Test Environment and Test Data Preparation
7. Requirement Traceability Matrix

6. Levels of Testing

1. Testing Techniques Used in Levels of Testing
2. White Box Testing Technique
3. Black Box Testing Technique
4. Unit Testing
5. Integration Testing
6. System Testing
7. Functional Testing
8. Non-Functional Testing
9. User Acceptance Testing

7. Types of Testing

1. Functional Testing
2. White Box Testing
3. Black Box Testing
4. Positive Testing
5. Negative Testing
6. Beta Testing
7. Live Environment Testing
8. Smoke Testing
9. Sanity Testing
10. Regression Testing

11. Formal Testing
12. Informal Testing
13. Monkey Testing
14. Re-Testing
15. Risk-Based Testing
16. Non-Functional Testing
 - Performance Testing
 - Load/Stress Testing
 - Usability Testing
 - Accessibility Testing
 - Security Testing

8. Effective Software Metrics

1. Test Case Productive Preparation
2. Test Execution Summary
3. Test Case Coverage
4. Defect Acceptance and Rejection
5. Test Efficiency
6. Effort Variance
7. Schedule Variance

9. Requirements Traceability Matrix (RTM)

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1. Forward Traceability
2. Backward Traceability
3. Bi-Directional Traceability
4. Examples of RTM

10. Testing Techniques

1. Equivalence Class Partition
2. State Transition Technique
3. Decision Table Technique
4. Boundary Value Analysis
5. Error Guessing Technique
6. Adhoc Testing



11. Defect Management

1. Difference Between Defect and Bug
2. Defect Life Cycle
3. Defect Tracking Tools
4. Create a Bug Report
5. Raise a Bug in JIRA
6. Severity and Priority

12. Agile Methodologies

1. Overview of Agile Methodologies

2. Agile Scrum
3. Agile Ceremonies
4. Agile Artifacts
5. Kanban
6. Adaptive Project Framework (APF)
7. Extreme Project Management (XPM)

13. Agile Scrum

1. Why Choose Agile?
2. Agile Manifesto and Scrum
3. Roles in Agile
4. Key Terminologies in Agile
5. Ceremonies in Agile
 - Sprint Grooming
 - Sprint Planning
 - Daily Scrum Meeting
 - Sprint Review
 - Sprint Retrospective

14. Testing Tools• JIRA

1. Introduction to JIRA and Test Management
2. JIRA Query Language (JQL)

3. Generating Reports in JIRA
4. JIRA Agile and Scrum

15. Live Testing Projects

16. Automation Testing Tools

1. Core Java
2. Selenium
3. TestNG
4. Data Driver
5. Cucumber
6. Maven

17. Interview Preparation

1. Manual Testing Interview Questions
2. Real-Time Interview Questions and Answers
3. Agile Testing Interview Questions
4. JIRA Interview Questions

18. Understanding Real-Time Scenarios and Work Culture

1. Project Explanation
2. Challenges Faced by Testers
3. Roles and Responsibilities of a Tester
4. Day-to-Day Activities of a Tester

19. Real-Time Projects

5. Work on two industry-based projects that discuss significant real-time use
6. cases aligned with the curriculum.

20. Resume Preparation

1. Creating an Impressive Resume

21. Mock Interview

- Mock Interviews with Industry Experts

